

Ethan Siazon

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EDUCATION

Texas Tech University, Lubbock, TX

Expected Graduation: May 2027

B.S. in Computer Engineering

4.0 GPA

Relevant Coursework: Data Structures & Algorithms, Circuits, Signals & Systems, Digital Logic, Electronics, Microcontrollers, Embedded Systems, Robotics, Software Engineering, Computer Networks, FPGA Design.

EXPERIENCE

Occidental Petroleum (OXY)

Houston, Texas

Software Engineering Platform Intern

May 2026 - August 2026

- Built and deployed three enterprise AI developer agents for repository risk analysis, AI-assisted software development, and automated code review, integrating them with GitHub Copilot and VS Code for enterprise-wide use.
- Reduced key engineering workflows from hours to minutes, cutting repository risk assessments from ~3.4 hours to ~5 minutes, application setup from ~5 hours to ~15 minutes, and code reviews from ~1 hour to ~3 minutes.
- Automated GitHub development workflows to analyze source code, identify security/compliance risks, enforce Oxy coding standards, and generate actionable issues and review comments linked directly to affected code.

X-Fab

Lubbock, Texas

Defect Detection Software Engineer Intern

January 2025 - December 2025

- Developed a Python and OpenCV-based computer vision system to detect and classify macro-level defects on SiC semiconductor wafers, accelerating defect analysis and ML dataset generation.
- Designed and implemented a web-based interface to streamline the configuration of detection parameters, enabling broader usability for engineers and process technicians.
- Contributed to the AI training pipeline by generating high-quality, labeled datasets used to train classification models.

PROJECTS

Automated Package Theft Detection System - Microcontrollers Project Lab

Embedded IoT Project

January 2026 - May 2026

- Designed and built an automated bait-package system using a Raspberry Pi 5, accelerometer, limit switches, GPS/4G, and four wide-angle cameras for real-time theft detection and tracking.
- Developed Python software to trigger recordings, upload evidence, send Discord location alerts, control mechanical responses, and stream live GPS/activity data to a custom tracking website.
- Integrated and tested electrical, mechanical, and software subsystems within a portable enclosure, successfully validating reliable event detection, remote monitoring, and full-system operation.

Remote-Controlled Battle Rover - Robotics Project Lab

Embedded Robotics Project

August 2025 - December 2025

- Designed and built a mobile battle rover using dual MSP430 microcontrollers, implementing PWM motor control, sensor interfacing, and IR-based wireless remote communication.
- Developed embedded C firmware for real-time motor control, navigation logic, and reliable wireless command processing.
- Integrated mechanical, electrical, and software subsystems into a fully functional, competition-ready robotic platform.

Tennis Analysis System

AI/ML Passion Project

November 2024

- Implemented a computer vision pipeline using YOLOv8 for real-time detection of tennis players and balls, fine-tuning on a custom dataset to improve accuracy.
- Developed and trained a convolutional neural network (CNN) using PyTorch to identify tennis court keypoints, enabling perspective correction and spatial analysis.

LEADERSHIP

Texas Tech Spikeball Club (Raider Roundnet)

Co-Founder & Secretary

September 2023 – Present

- Co-founded the club by drafting its constitution and bylaws, coordinating with faculty advisors, securing formal TTU recognition, and helping establish the club's organizational structure and leadership processes.
- Drove membership growth through social media, tabling, and campus events while developing onboarding and communication processes to improve retention, engagement, and overall member involvement.
- Managed operations for membership rosters, officer elections, scheduling, and communications for officers and members.